Computer Graphics

Arbitrary Rotations

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Two Dimensional Rigid Body Transform

If a transformation matrix includes only translation and rotation parameters, it is a **rigid-body transformation matrix.** The general form for a two-dimensional rigid-body transformation matrix is

$$\begin{bmatrix} r_{xx} & r_{xy} & tr_x \\ r_{yx} & r_{yy} & tr_y \\ 0 & 0 & 1 \end{bmatrix} \tag{45}$$

where the four elements r_{jk} are the multiplicative rotation terms, and the elements tr_x and tr_y are the translational terms. A rigid-body change in coordinate position is also sometimes referred to as a **rigid-motion** transformation. All angles and distances between coordinate positions are unchanged by the transformation. In addition, matrix 45 has the property that its upper-left 2×2 submatrix is an *orthogonal matrix*. This means that if we consider each row (or each column) of the submatrix as a vector, then the two row vectors (r_{xx}, r_{xy}) and (r_{yx}, r_{yy}) (or the two column vectors) form an orthogonal set of unit vectors. Such a set of vectors is also referred to as an *orthonormal* vector set. Each vector has unit length as follows:

$$r_{xx}^2 + r_{xy}^2 = r_{yx}^2 + r_{yy}^2 = 1 \tag{46}$$

and the vectors are perpendicular (their dot product is 0):

$$r_{xx}r_{yx} + r_{xy}r_{yy} = 0 \tag{47}$$

Therefore, if these unit vectors are transformed by the rotation submatrix, then the vector (r_{xx}, r_{xy}) is converted to a unit vector along the x axis and the vector (r_{yx}, r_{yy}) is transformed into a unit vector along the y axis of the coordinate system:

$$\begin{bmatrix} r_{xx} & r_{xy} & 0 \\ r_{yx} & r_{yy} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{xx} \\ r_{xy} \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \tag{48}$$

$$\begin{bmatrix} r_{xx} & r_{xy} & 0 \\ r_{yx} & r_{yy} & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} r_{yx} \\ r_{yy} \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} \tag{49}$$

Computer Graphics with OpenGL (4th ed.) [Hearn, Baker & Carithers 2013]

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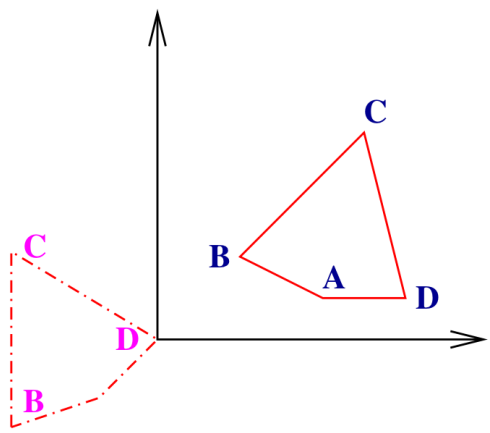
A Transformation Problem

- Given Position of B, C and D only
- Bring D to origin and BC parallel to the Y axis as shown by the dotted polygon



Transformation Computation

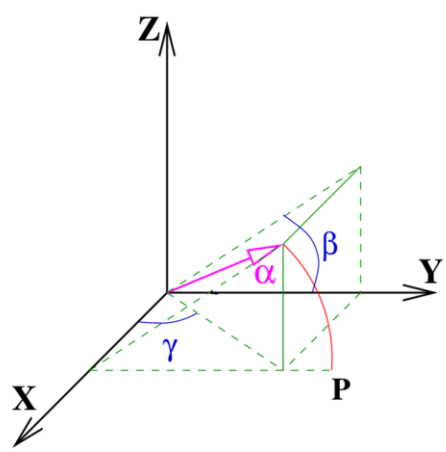
1. Translate by $-D$
   - What is the orientation of BC ?
   - It has be made parallel to Y axis
2. Create a unit vector $\vec{v} = [v_x \quad v_y]^T$ representing the orientation from B to C
   - This vector would be the second row of the Rotation Matrix
   - $R = \begin{bmatrix} u_x & u_y \\ v_x & v_y \end{bmatrix}$
   - $\vec{u} = [u_x \quad u_y]^T$ is orthogonal to $\vec{v}$ and can be represented using $\vec{v}$
   - $\vec{u}$ can be computed by rotating $\vec{v}$ by R_θ with $\theta = -90$, $\vec{u} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} v_y \\ -v_x \end{bmatrix}$
3. $R = \begin{bmatrix} \vec{u} \\ \vec{v} \end{bmatrix} = \begin{bmatrix} v_y & -v_x \\ v_x & v_y \end{bmatrix}$
4. The complete transformation $M = T(-D)R$

Rotation about an axis parallel to Z

- An axis parallel to Z axis, passing through point $(x, y, 0)$.
- Translate so that the axis passes through the origin: $T(-x, -y, k)$ for any k
- Overall: $M = T(x, y, -k)\, R_Z(\theta)\, T(-x, -y, k)$
- Why shouldn't k matter?
- $R_Z(\theta)$ doesn't affect the Z coordinate. So, whatever k is added first will be subtracted later

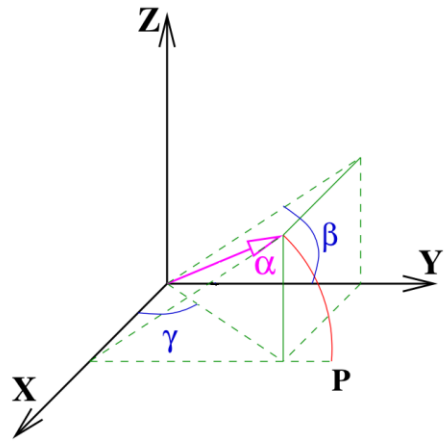
Rotation θ about an axis α

- $\mathbf{R}_\alpha(\theta)$
- 3-step process:
  - Apply $\mathbf{R}_{\alpha\mathbf{x}}$ to align α with the X axis
  - Rotate about X by angle θ
  - Undo the first rotation using $\mathbf{R}_{\alpha\mathbf{x}}^{-1}$
- $\mathbf{R}_\alpha(\theta) = \mathbf{R}_{\alpha\mathbf{x}}^{-1}\, \mathbf{R}_\mathbf{x}(\theta)\, \mathbf{R}_{\alpha\mathbf{x}}$
- What is $\mathbf{R}_{\alpha\mathbf{x}}(\theta)$?
- We can align α with Y or Z axis also



Rotation θ about an axis α

- First rotate by $-\beta$ about $\mathbf{X}$ axis, $\mathbf{R_x}(-\beta)$
- Vector α would lie in the $\mathbf{XY}$ Plane
- Then rotate by $-\gamma$ about $\mathbf{Z}$ axis, $\mathbf{R_z}(-\gamma)$
- Vector α would coincide with the $\mathbf{X}$ axis
- $\mathbf{R}_{\alpha\mathbf{x}} = \mathbf{R_z}(-\gamma)\mathbf{R_x}(-\beta)$
- Then rotate by θ about $\mathbf{X}$ axis, $\mathbf{R_x}(\theta)$
- $\mathbf{R}_{\alpha}(\theta) = \mathbf{R}_{\alpha\mathbf{x}}^{-1}\,\mathbf{R_x}(\theta)\,\mathbf{R}_{\alpha\mathbf{x}}$
- Since $\mathbf{R}_{\alpha\mathbf{x}}^{-1} = \mathbf{R_x}(\beta)\mathbf{R_z}(\gamma)$
- $\mathbf{R}_{\alpha}(\theta) = \mathbf{R_x}(\beta)\mathbf{R_z}(\gamma)\,\mathbf{R_x}(\theta)\,\mathbf{R_z}(-\gamma)\mathbf{R_x}(-\beta)$



Alternative for $R_{\alpha x}$

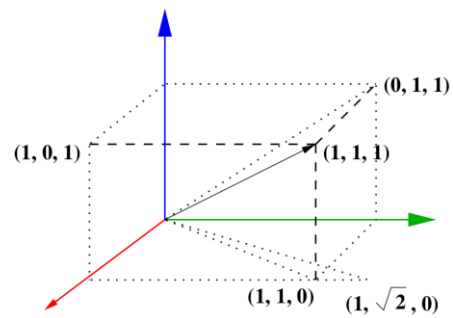
- Don't we know about rotation matrices and direction cosines that go to/from coordinate axes?
- $R_{\alpha x} = \begin{bmatrix} \mathbf{r_1} & 0 \\ \mathbf{r_2} & 0 \\ \mathbf{r_3} & 0 \end{bmatrix}$
- $R_{\alpha x} = \begin{bmatrix} \alpha & 0 \\ \alpha \times \mathbf{v} & 0 \\ \mathbf{r_1} \times \mathbf{r_2} & 0 \end{bmatrix}$

- After rotation, α will align with $\mathbf{X}$-axis. Hence that is the first row $\mathbf{r_1}$ of the rotation matrix
- Find a direction orthogonal to α to be row $\mathbf{r_2}$. Take any vector v not parallel to α. $\mathbf{r_2} = \alpha \times \mathbf{v}$
- Finally, $\mathbf{r_3} = \mathbf{r_1} \times \mathbf{r_2}$
- $\mathbf{r_2}$ and $\mathbf{r}_3$ would depend on $\mathbf{v}$ and might create different basis vectors. However in the final computation of $\mathbf{R}_{\alpha}(\theta)$, the variation in $\mathbf{R}_{\alpha\mathbf{x}}$ is negated by its inverse computation

Rotation $-\frac{\pi}{4}$ about an axis $\alpha = [1,1,1]^{\mathrm{T}}$

Method-1

- $\mathbf{R}_{\alpha\mathbf{x}} = \mathbf{R}_{\mathbf{z}}(-\gamma)\mathbf{R}_{\mathbf{x}}(-\beta)$
- $\mathbf{R}_{\mathbf{x}}(-\beta) = \mathbf{R}_{\mathbf{x}}(-\arctan(1))$
- $\mathbf{R}_{\mathbf{x}}(-\beta) = \mathbf{R}_{\mathbf{x}}\left(-\frac{\pi}{4}\right)$
- $\mathbf{R}_{\mathbf{x}}(-\beta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
- $\mathbf{R}_{\mathbf{z}}(-\gamma) = \mathbf{R}_{\mathbf{z}}(-\arctan\sqrt{2})$
- $\mathbf{R}_{\mathbf{z}}(-\gamma) = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & 1 \end{bmatrix}$



$\tan\beta = 1, \qquad \tan\gamma = \sqrt{2}$

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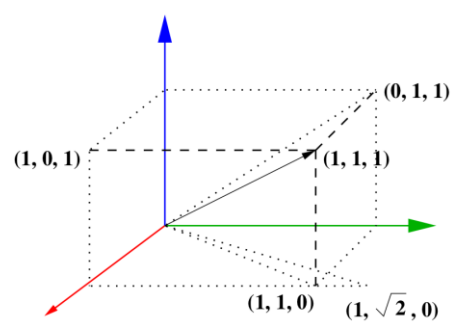
Rotation $-\frac{\pi}{4}$ about an axis $\alpha = [1,1,1]^{\mathrm{T}}$

Method-1

- $\mathbf{R}_{\alpha\mathbf{x}} = \mathbf{R}_{\mathbf{z}}(-\arctan\sqrt{2})\mathbf{R}_{\mathbf{x}}(-\arctan(1))$
- $\mathbf{R}_{\alpha\mathbf{x}} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} & 0 \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$
- $\mathbf{R}_{\alpha\mathbf{x}} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{6}} \\ 0 & -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$



$\tan\beta = 1, \qquad \tan\gamma = \sqrt{2}$

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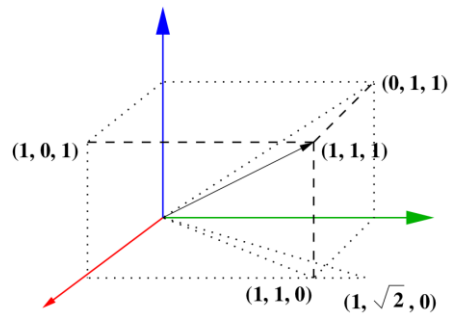
Rotation $-\frac{\pi}{4}$ about an axis $\alpha = [1,1,1]^{\mathrm{T}}$

Method-2

- $[1,1,1]^{\mathrm{T}}$ will lie on X-axis
- $\mathbf{r_1} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \end{bmatrix}^{\mathrm{T}}$
- $\mathbf{r}_2 = \mathbf{r_1} \times [1 \quad 0 \quad 0]^{\mathrm{T}} = \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}^{\mathrm{T}}$
- $\mathbf{r}_3 = \mathbf{r_1} \times \begin{bmatrix} 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{bmatrix}^{\mathrm{T}} = \begin{bmatrix} \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix}^{\mathrm{T}}$
- $\mathbf{R}_{\alpha\mathbf{x}} = \begin{bmatrix} \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$



Rotation Around Arbitrary Axis

- An arbitrary axis may not pass through the origin
- Translate by $\mathbf{T}$ so that it passes through the origin
- Apply $\mathbf{R}_\alpha$
- Translate back using $\mathbf{T}$
- Composite transformation: $\mathbf{T}^{-1}\mathbf{R}_\alpha\mathbf{T}$

Thank You

Quiz-I Tomorrow